

Austin Senna Wijaya

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EDUCATION

Columbia University

Bachelor of Science: Computer Science

New York, NY

Aug 2024 – May 2028

- GPA: 4.22/4.0; Full-Tuition Scholarship; Dean's List.
- Research interests: data systems, AI agents, retrieval systems, machine learning.

RESEARCH EXPERIENCE

AI Research Assistant

Dec 2025 – Present

Columbia Data, Agents, and Processes Lab

New York, NY

- Engineered a high-throughput benchmark evaluation harness in **Strands SDK**, using parallel sandboxed **DuckDB/Python** runtimes for multi-agent question-answering tasks incorporating skills, hooks, and context compaction.
- Processed **9.5 TB / 40M docs** into an S3 data lake; built scalable **BM25** and **vector search** in LanceDB for massive-scale data retrieval for agents.
- Architected telemetry to capture tool-call traces and track agent reasoning to identify bottlenecks in agent framework, improving benchmark accuracy for GPT-5-mini from **5.16%** to **56.3%**.

Machine Learning Research Assistant

Dec 2025 – May 2026

Columbia Zuckerman Institute

New York, NY

- Built and parallelized a GPU-optimized **PyTorch** pipeline in HPC clusters for **Qwen2-Audio** inference, hidden-state extraction, and supervised classification training under cross-validation across 12 speech attributes and **120K+ audios**.
- Curated training-ready audio datasets from 12 speech and language benchmarks totalling 120K+ audios, including converting text benchmarks into audio through parallel **Kokoro-TTS** generation.
- Trained bottleneck FNNs to compress **4096-D** hidden-state embeddings into **10-100D** feature-specific representations, isolating individual linguistic features from **Qwen2-Audio** for downstream analysis.

PUBLICATIONS

- [1] *LakeQA: A Benchmark for Complex Exploratory QA over a Million-Scale Data Lake*, **ICML 2026**: Haonan Wang, Jiaxiang Liu, Yurong Liu, **Austin Senna Wijaya**, Tianle Zhou, Eden Wu, Yijia Chen, Wanting You, Reya Vir, Daniela Pinto Veizaga, Grace Fan, Yusen Zhang, Juliana Freire, Eugene Wu.
- [2] *SANA: What Matters for QA Agents over Massive Data Lakes?* **VLDB DASHSys 2026**: **Austin Senna Wijaya**, Jiaxiang Liu, Haonan Wang, Eugene Wu.
- [3] *MINA: Linear Probes Reveal Coding-Sequence Family Signal in Frozen DNA Encoders*, **PMLR MLCB 2026**: **Austin Senna Wijaya**, Hayden Leung.
- [4] *Targeted Embeddings: An Interpretable Framework for Neural Speech Encoding with Large Audio-Language Models*, **NeurIPS 2026**: Linyang He, Ciro D. Randazzo, **Austin Senna Wijaya**, Gabriel Diaz, Yunyan Ouyang, Jingting Zhang, Jonathan Brennan, Nima Mesgarani. (*Under-review*).
- [5] *Be CARE-ful with Text-to-SQL Benchmarks*, **NeurIPS 2026**: Haonan Wang, Jiaxiang Liu, Elaine Ang, Tiancheng Ge, Wanting You, **Austin Senna Wijaya**, Reya Vir, Tianle Zhou, Mihir Agarwal, Yusen Zhang, Eugene Wu. (*Under-review*).

INDUSTRY EXPERIENCE

Software Engineer Intern

Jun 2026 – Aug 2026

Garner Health

New York, NY

- Engineered observability stack for newly launched B2B web portal, integrating **Datadog**, **OpenTelemetry**, and **Grafana** to establish end-to-end distributed tracing across frontend and backend services, surfacing previously invisible user-impacting bugs and increasing the team's bug resolution throughput by 3x.
- Automated telemetry provisioning, alerting, and monitoring infrastructure across Grafana and Datadog using **Terraform**, **GitLab CI/CD**, and **Helm**, replacing manual observability setup with reusable, version-controlled configuration.
- Deployed a self-improving autonomous on-call triage agent on **Kubernetes** using **Claude Agent SDK** with MCP tool access (PostgreSQL, Playwright, Jira, GitLab) to generate proactive alerts from telemetry data, investigate incident root cause, and ship fix pull requests; autonomously resolved **15+ production incidents**.

Data Engineering Intern

Jun 2025 – Aug 2025

Ruangguru (Indonesia's largest edtech platform)

Jakarta, Indonesia

- Built scalable ETL pipelines in **BigQuery** to transform millions of telemetry events into structured session tables, mapping end-to-end payment conversion funnel to pinpoint drop-off points.
- Architected an automated web scraping, data ingestion, and sentiment analysis service utilizing **Puppeteer** and **Hugging Face** APIs, surfacing location review intelligence on live **Looker Studio** dashboards.

TECHNICAL SKILLS

- **Languages:** Python, SQL, JavaScript / TypeScript, Java, C
- **AI & Data:** Strands Agents, Claude Agent SDK, Langchain, Pandas, PyTorch, Playwright, BeautifulSoup
- **Cloud & Infra:** AWS (EC2, S3, Bedrock), GCP (BigQuery, GKE), Grafana, OpenTelemetry, Kubernetes, Terraform, Docker
- **Framework & Tools:** Next.js, React, FastAPI, Litestar, Electron, GitHub, Claude Code

HONORS

- **Databricks Grounded Reasoning Cup 2026:** Columbia representative; worked with Google.
- **Venture Competitions 2026:** 2nd Place at Columbia Lion Cage; Top 6 at Millard Chan Technology Startup Competition.
- **Hack@Brown 2026:** Finalist; Strongest Product Thinking award; worked with Google Ventures and Partiful.
- **Jane Street Mystery Planet Trading & Puzzle Competition 2026:** 1st Place.
- **Tencent Kepler Plan S3 Competition 2025:** 2nd Place; Best Code; Best Product Idea; Popularity Award.
- **55th International Chemistry Olympiad 2023:** Final stage, 10th rank for Indonesian national team selection.